

Density Scene-Fitting Specification

Closed-loop adaptation of periodic HDR shells to named browser profiles

mdstats

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Purpose and status

Module: `mdstats.plotting.density_scene_fit`

Status: normative and implemented through `mdstats 0.19.78a0`.

This module owns display-complexity adaptation after scientific scalar fields and initial meshes exist. It replaces estimate-and-abort behavior with an exact closed-loop controller.

Ownership boundary

The module owns:

- `BrowserMeshProfile`;
- backend-neutral `DensityShellGeometry`;
- shell and scene fit-attempt/result records;
- weighted target reallocation;
- bounded simplification retries;
- lower-resolution recontouring of the same scalar field;
- exact post-replication verification against `BrowserMeshBudget`.

It does not own density estimation, HDR threshold selection, QEM internals, raw mesh extraction, or Plotly serialization.

Browser profiles

Profile	Final density faces	Final density vertices	HTML bytes	Plotly traces
compact	300,000	225,000	40 MiB	72
balanced	600,000	450,000	72 MiB	96
quality	1,000,000	750,000	128 MiB	128
custom	caller supplied	caller supplied	caller supplied	caller supplied

balanced is the default. These are conservative package presets, not universal browser or GPU guarantees.

Validated extraction prerequisite

The scene fitter accepts only periodic meshes that already satisfy canonical seam pairing and edge-incidence requirements. Sparse extraction validates the globally welded tile surface before it creates `DensityShellGeometry`. Invalid tile-local presimplification is retried without local reduction; persistent extraction defects are repaired by bounded coarse recontouring or the declared non-mesh fallback.

The fitter therefore adapts display complexity, not malformed topology. Every QEM or recontour candidate produced inside this module is validated again before acceptance.

Backend-neutral shell geometry

```
DensityShellGeometry(  
    shell_key: str,  
    field: ScalarField3D,  
    mass_fraction: float,  
    contour_level: float,  
    vertices_fractional: ndarray[n, 3],  
    vertices_cartesian: ndarray[n, 3],  
    faces: ndarray[m, 3],  
    display_replication: int,  
    visual_importance: float,  
    minimum_faces: int,  
    source_kind: str,  
)
```

Dense and sparse contour paths must convert to this representation before fitting. Plotly objects must not enter the fitting layer.

Scientific invariants

Fitting may alter display geometry but must not alter:

- selected frames or weights;
- field values or normalization;
- Gaussian width or kernel cutoff;
- scientific logical grid;
- requested HDR mass fraction;
- scientific contour threshold;
- display cell or periodicity;
- topology-category identity.

Recontouring samples the already prepared scalar field at a coarser display grid and uses the same contour level. It is not a new density estimate.

Closed-loop algorithm

For shell s with canonical faces F_s and display multiplicity m_s , final compliance requires

$$\sum_s m_s F_s \leq F_{\max}$$

with simultaneous vertex, trace, and HTML constraints.

The controller performs:

1. exact measurement of current post-replication usage;
2. initial target allocation from `density_scene_budget`;
3. periodic QEM simplification under strict fidelity limits;
4. browser-profile QEM retry with bounded relaxation;
5. compensated numerical targets when the simplifier overshoots;
6. lower-resolution recontouring when permitted;

7. periodic seam, triangle-incidence, finite-value, and degeneracy validation;
8. exact final budget evaluation before Plotly assembly.

Inner HDR shells receive greater visual importance than diffuse outer shells. The outer shell therefore absorbs more reduction when an additional scene-wide reduction is required.

Fit reports

DensityShellFitAttempt
DensityShellFitResult
DensitySceneFitReport

Every accepted or rejected attempt records method, requested target, resulting faces and vertices, validation status, and diagnostic metadata. An irreducible failure raises `BrowserMeshBudgetFailure` containing the scene report.

Failure policy

Failure is reserved for cases where no permitted periodic candidate can satisfy the selected profile. The controller must not fail merely because:

- the first simplification misses its target;
- one shell initially exceeds 250,000 faces in scene-controller mode;
- the initial scene is slightly above its profile budget.

The historical 301,838-, 314,640-, and 582,375-face cases must enter this controller.

Optional dependencies

QEM retries require `fast-simplification`. Recontouring requires `scikit-image`. Both are included by `mdstats[interactive]`. Missing optional dependencies must produce an explicit capability failure or a remaining permitted fallback; they must not silently skip final budget validation.

Focused validation

Tests must cover:

- profile coercion and custom budgets;
- dense and sparse shell normalization;
- target misses routed into fitting;
- exact post-replication face and vertex accounting;
- QEM retry and overshoot compensation;
- recontour fallback using an unchanged threshold;
- seam and incidence validation after every accepted candidate;
- deterministic repeated output;
- structured irreducible failure;
- the three historical count regressions.

External method

Mesh reduction uses quadric-error metrics following Garland and Heckbert, “Surface Simplification Using Quadric Error Metrics,” SIGGRAPH 1997, DOI: 10.1145/258734.258849. Contour extraction uses the Lewiner et al. topologically consistent marching-cubes method, *Journal of Graphics Tools* 8(2), 1–15 (2003), DOI: 10.1080/10867651.2003.10487582.

The weighted scene controller, retry ladder, periodic candidate validation, and profile policy are `mdstats`-specific integration designs.